

Effects Of NMES On EMG-EMG Coherence Across Glenohumeral Muscles During Fatigue In Individuals With Rotator Cuff Disease

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Introduction

- Rotator cuff disease (RCD) causes anterolateral shoulder pain.
- Electromyographic (EMG-EMG) coherence quantifies the shared oscillatory inputs to motoneuron pools between a pair of muscles in the frequency domain.
- Neuromuscular electrical stimulation (NMES) can enhance neuromuscular control via motor and sensory neural pathways.

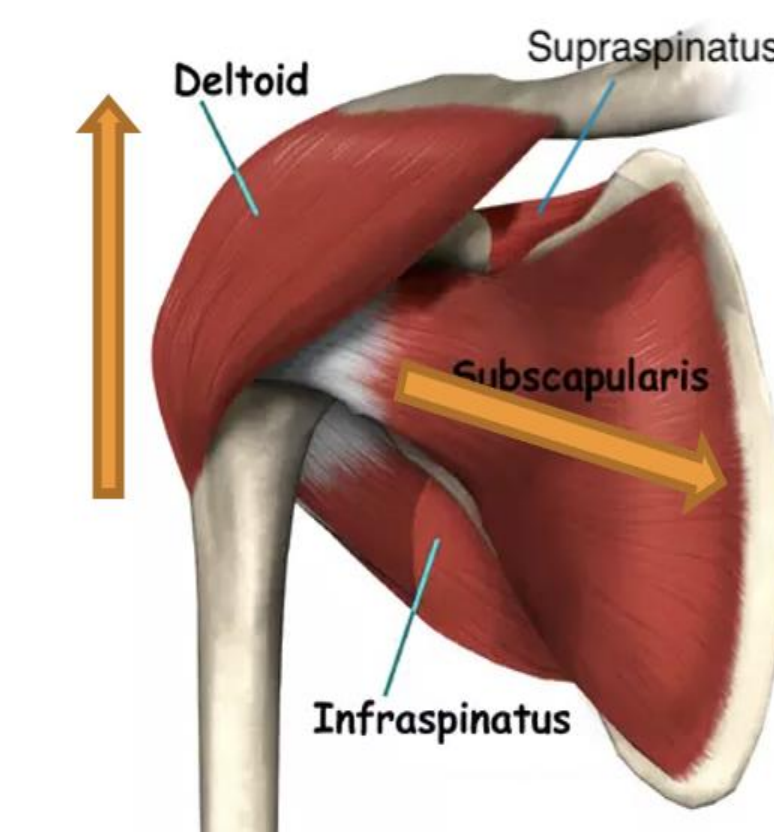


Fig 1. Force during abduction

Purpose

To determine if NMES can improve EMG-EMG coherence across the rotator cuff and deltoid muscles during fatigue in RCD.

Methods

- Participants: 8 adults with RCD (39 ± 19 yrs)
- Intramuscular EMG: supraspinatus (SS); sEMG: infraspinatus (IS), middle deltoid (MD).
- Fatiguing Task: 25% MVC in 30° scapula plane until endurance limit.
- Interventions: NMES on SS (anode) & IS (cathode) at maximal tolerable intensity (40 Hz; 0.46ms pulse; 4s-on, 6s-off); Sham at 0 mA.
- Statistical Analysis: Coherence in the delta (2-5 Hz) and beta (15-35 Hz) bands were compared using a two-way RM ANOVA (Time $\times$ Treatment) with Bonferroni post-hoc analysis.



Fig 2. NEUBIE stimulator.

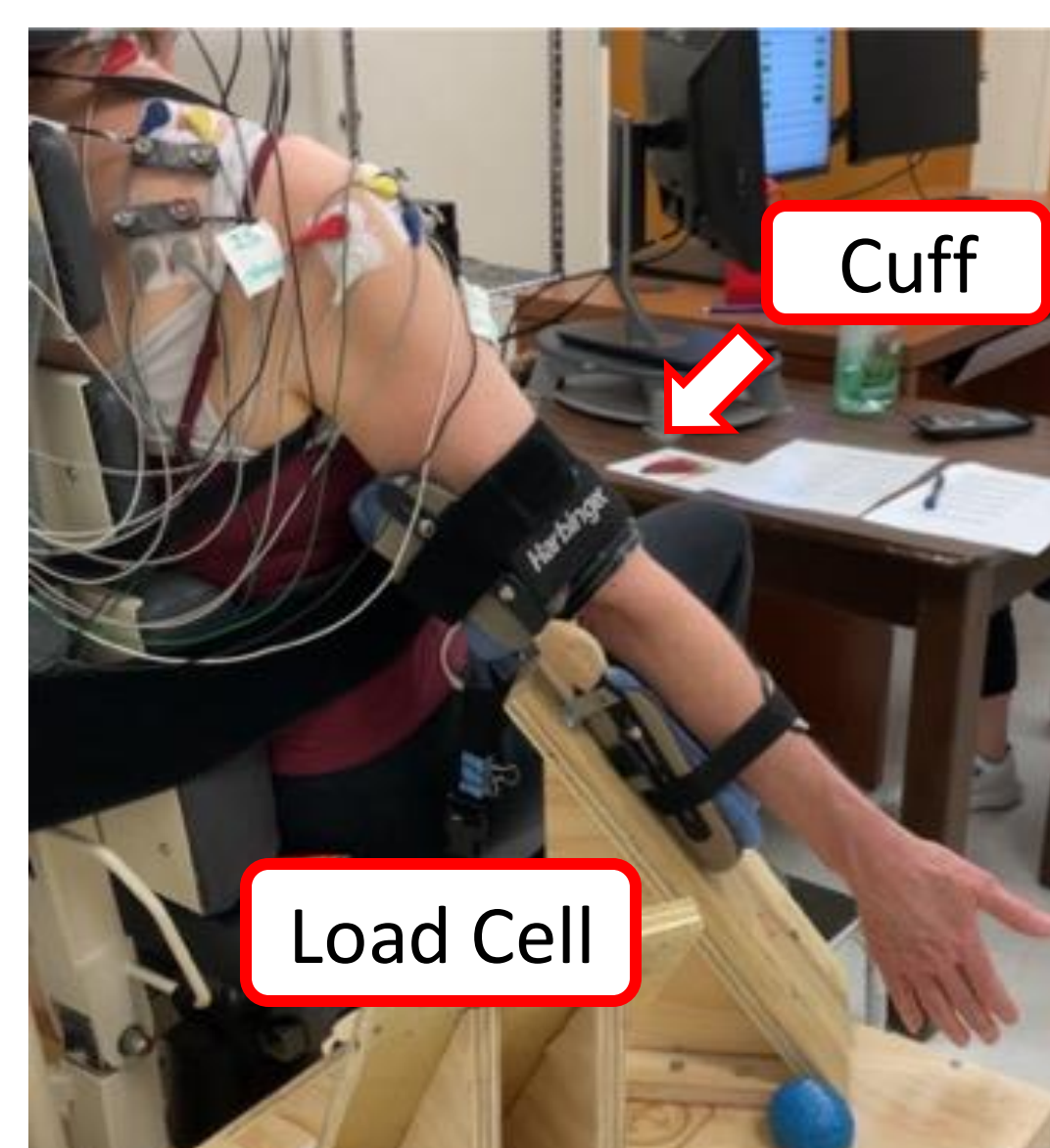


Fig 3. Experimental setting.

Methods

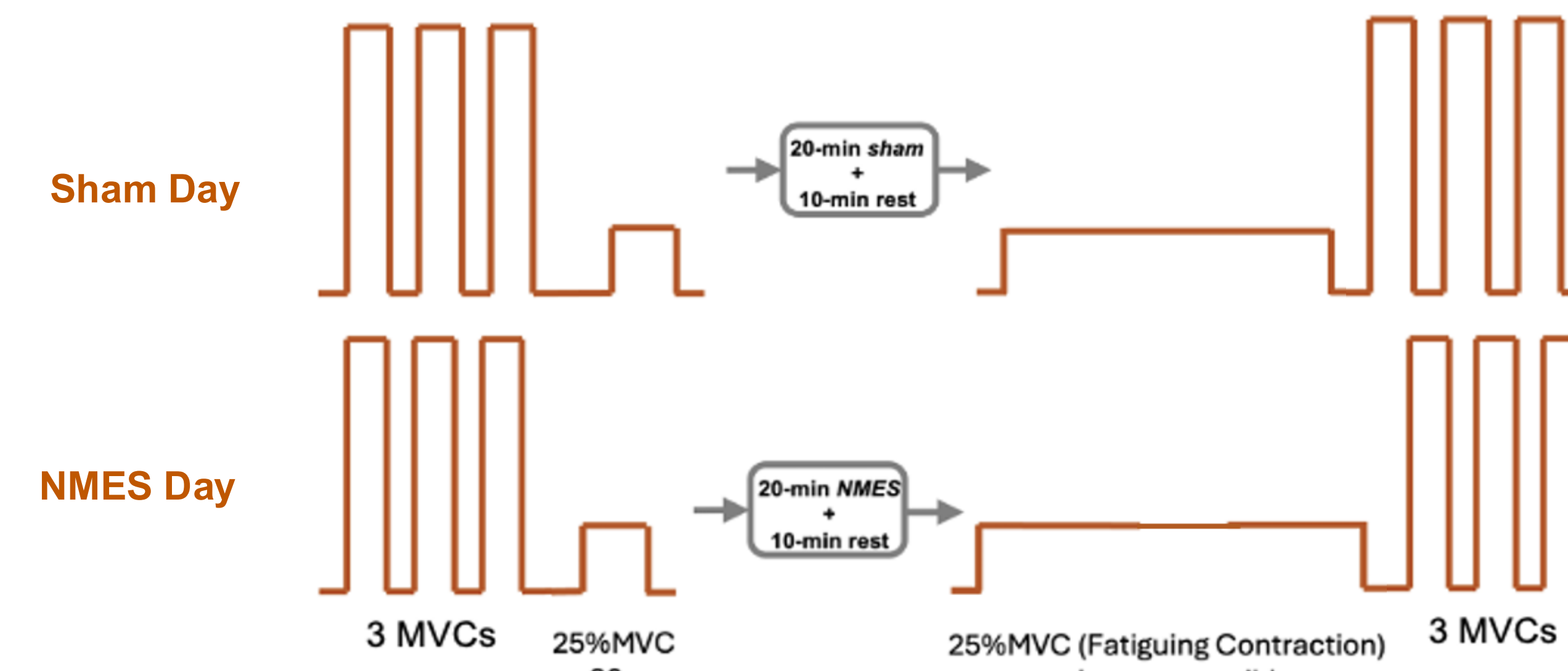


Fig 4. Experimental protocol.

Results

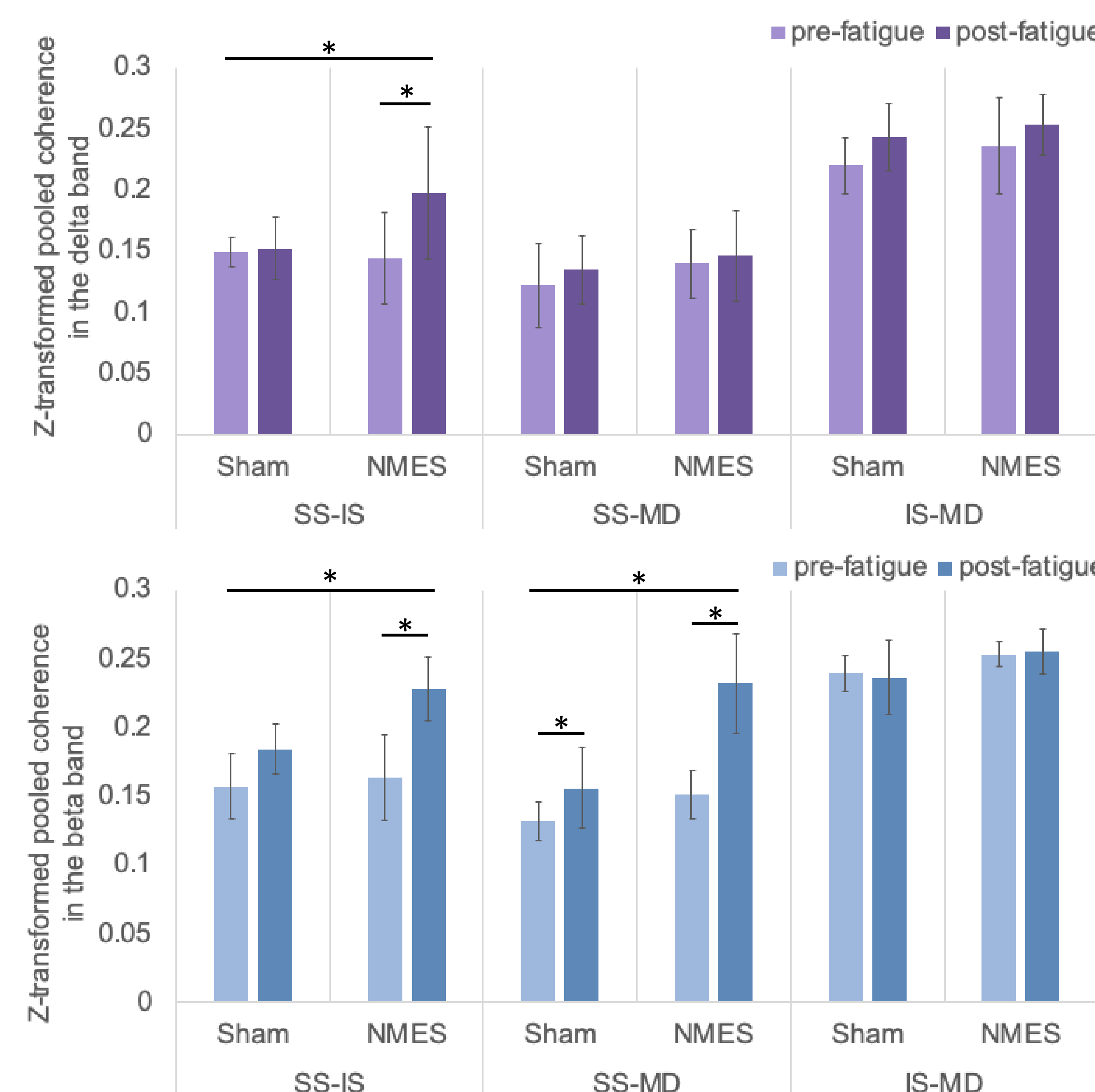


Fig 5. Z-transformed pooled coherence in delta and beta frequency bands, * $p < 0.05$.

- Baseline Coherence:** No significant differences in any bands between the two testing days.
- Delta-Band Coherence:** A significant increase during fatigue ($p = 0.002$) was observed in the SS-IS pair after NMES.
- Beta-Band Coherence** increased in both conditions during fatigue. A greater increase was observed after NMES for SS-IS ($p = 0.04$) and SS-MD ($p = 0.02$) than after sham.
- No significant difference was observed in IS-MD coherence between NMES and sham conditions.

Results

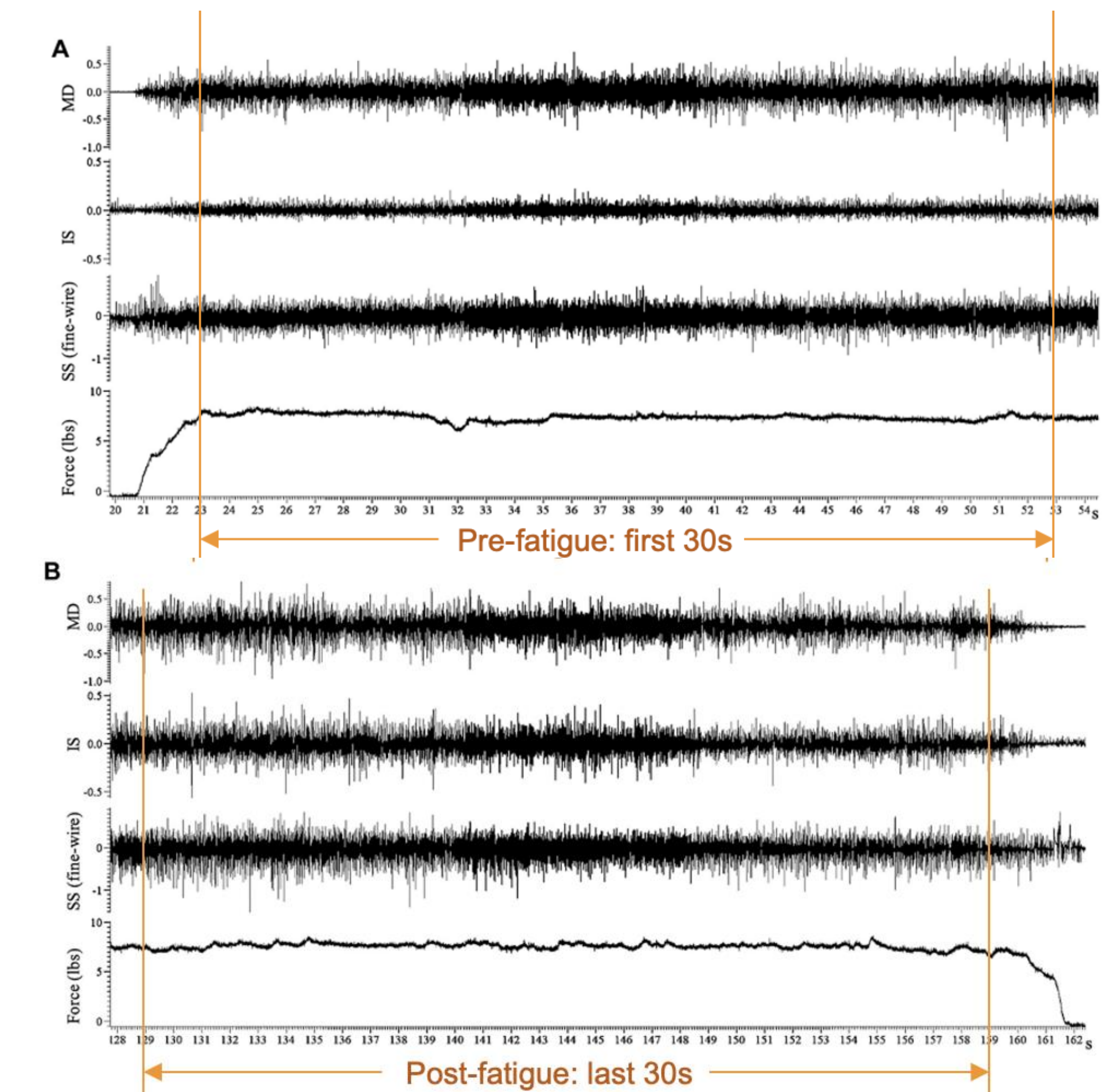


Fig 6. Raw EMG data sample.

Conclusion

- NMES enhanced SS-IS and SS-MD coherence in the delta and beta bands during fatigue, suggesting increased shared neural drive between muscles, potentially reflecting greater excitability of the corticospinal pathway [1].
- The improved EMG-EMG coherence between antagonist muscles may contribute to enhanced rotator cuff activation and better shoulder joint stability under fatigue [2].
- The use of NMES as a non-invasive treatment to improve intermuscular coordination across glenohumeral muscles in RCD rehabilitation [3].

References

- [1] Kirkwood, P. A. (2016) J Neurophysiol, 115(2), 1077-1078.
- [2] Blickenstorfer, A., et al. (2009) Hum Brain Mapp, 30(3), 963-975.
- [3] Granovsky, Y., Sprecher, E., & Sinai, A. (2019). Pain Rep, 4(2), e725.